

Indonesia Egg Price Forecasting

A regime-aware approach to market volatility and calendar risk

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Final Project · Student Bootcamp Offline Data Analytics · Dibimbing Batch #3



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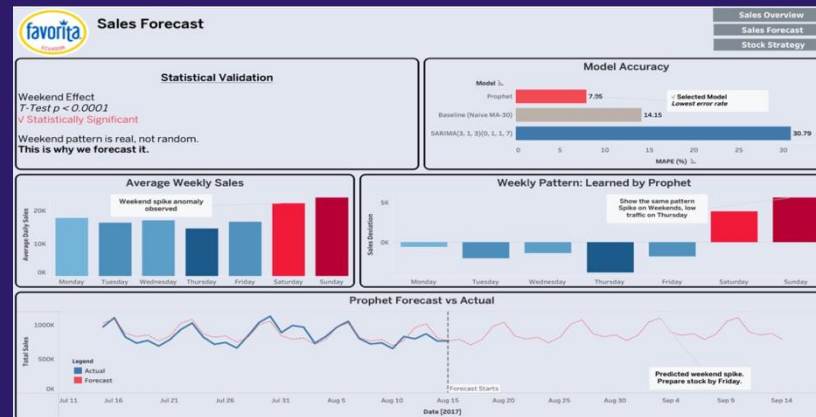
Retail Time Series Analysis

Tools: Python, Tableau

Problem: Ecuadorian retailer Corporación Favorita needed to forecast daily sales per product category to plan stock accurately and avoid over/understocking.

Result: Discovered a statistically significant weekend sales anomaly across all product categories and built a Prophet forecasting model that successfully predicted the recurring weekly spike pattern.

Recommendation: Prophet Stock preparation timing, category-specific inventory prioritization, and promotion calendar redesign based on data-driven demand patterns rather than assumptions.



2

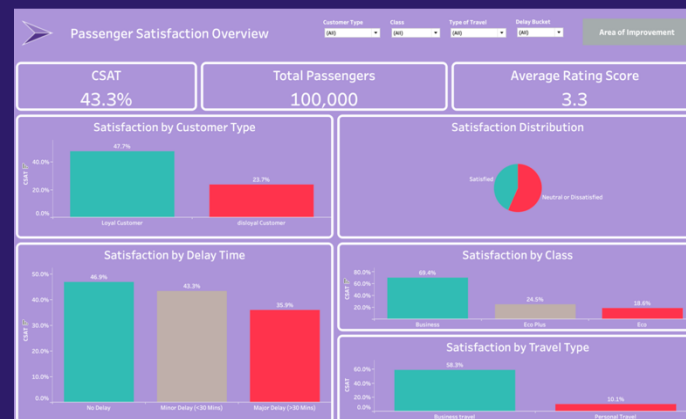
Airline Satisfaction Analysis

Tools: Python, Tableau

Problem: A low-cost airline needed to understand which service touchpoints most influence passenger satisfaction.

Result: Online boarding emerged as the strongest driver of satisfaction.

Recommendation: Prioritize the boarding experience to lift overall CSAT.



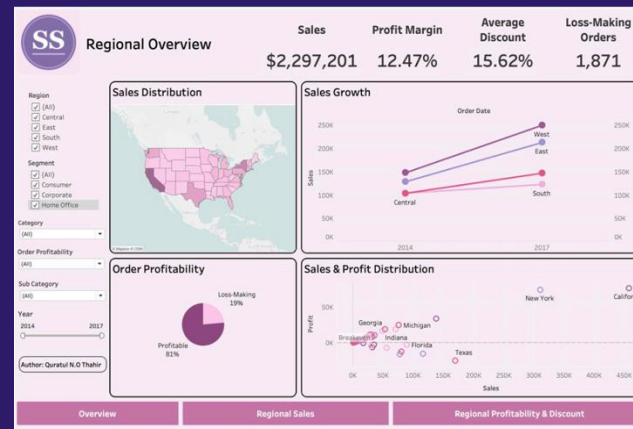
Tools: Python, Tableau

Recommendation : Audit post engagement and trial campaigns in underperforming regions.



Tools: Python, Tableau

Recommendation : Tighter discount caps for the Central and South regions.



Project Overview

5

Olist E-Commerce

Tools: SQL

Problem: Olist wanted to understand how customer geography shapes shopping patterns and delivery performance across Brazil.

Result: Northeastern Brazil showed strong demand paired with weak delivery performance.

Recommendation : Expand logistics coverage there and introduce a loyalty program to protect margins

6

Student Bags Analysis

Tools: Python

Problem: A bag retailer assumed waterproofing and laptop-compartment features justified a price premium and wanted this tested before using it in pricing strategy.

Result: The price difference was under \$1 and not statistically significant ($p > 0.05$)

Recommendation : Pair this dataset with real sales data before using it for pricing decisions.

KETEPATAN WAKTU PENGIRIMAN BERDASARKAN LOKASI

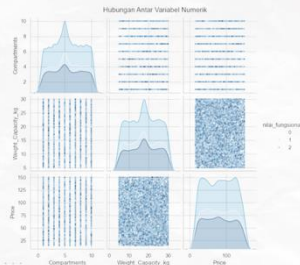
customer_state	total_orders	on_time_orders	late_orders	late_percentage
SP	40489	38103	2386	5.89
RJ	12351	10687	1664	13.47
MG	11352	10714	638	5.62
BA	3256	2799	457	14.04
RS	5342	4960	382	7.15
SC	3547	3201	346	9.75
PR	4923	4677	246	5.00
ES	1595	1751	244	12.23
CE	1278	1082	196	15.34
PE	1593	1421	172	10.80

Keterlambatan pengiriman tertinggi di CE (15.3%) dan BA (14%). Kedua lokasi berada di wilayah timur laut Brazil yang lebih jauh dari pusat distribusi.

RJ memiliki tingkat keterlambatan yang cukup tinggi (13.5%) meskipun merupakan kota besar.

Mengindikasikan adanya tantangan logistik di wilayah timur laut Brazil.

HUBUNGAN ANTAR VARIABEL



PEMILIHAN CHART

Menampilkan scatter plot untuk setiap kombinasi pasangan variabel sekaligus dalam satu chart (analisis multivariate)

nilai_fungsional sebagai hue: melihat apakah kelengkapan fitur fungsional memisahkan kelompok secara visual.

INTERPRETASI

Tidak ada pola hubungan yang terlihat di antara variabel (titik tersebar acak).

Tidak ditemukan hubungan signifikan antar variabel.

Project Overview

7

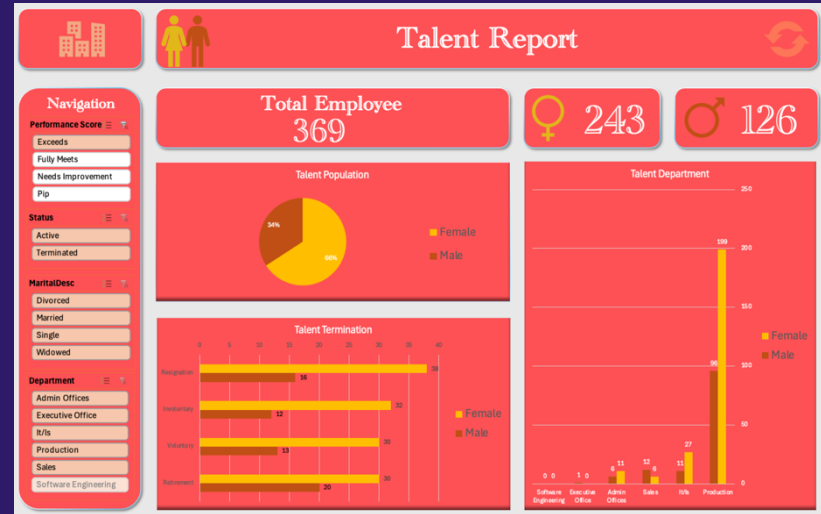
HR Workforce Analysis

Tools: Excel

Problem: Leadership wanted to know whether top talent was being placed and promoted where it could have the most strategic impact.

Result: Top performers (top 10%) were concentrated in Production rather than strategic roles, and 0% of executives were female despite 66% of top performers being female.

Recommendation : Targeted STEM recruitment and transparent promotion paths



Background and Problem Statement

Background

Eggs sit at the center of Indonesia's food security picture. Affordable, nutrient-dense, and consumed across every income bracket. Yet prices don't move steadily; they swing sharply around Ramadan, national holidays, and supply shocks, each capable of straining household budgets exactly when demand is highest. This volatility isn't just a market inconvenience. It's a direct challenge to the kind of food resilience **SDG 2: Zero Hunger** calls for. Since many of these demand-heavy events are predictable in timing. A well-built forecast offers something rare: the chance to prepare ahead of a price shock, rather than absorb it after the fact.



PROBLEM STATEMENT

Can Indonesia's egg prices be forecast reliably enough to support procurement and policy decisions?

Sub-Questions:

1. What does Indonesia's egg price behavior look like?
2. How much of price volatility can a statistical model actually predict?
3. What happens when the forecast is tested against a real, unseen outcome?

Data Understanding

WFP Global Food Prices dataset | Source: World Food Programme (HDX) [link](#)

Collection method: Officers were deployed to gather direct records from the field

1,847,522

Rows Observations

16

Columns / Features

4

Years
2022–2025

Key Columns

Dimension

date

Monthly price observation date

Dimension

market

Market name (province/city level)

Dimension

commodity

Commodity type

Measure

usdprice

Price per unit in USD

countryiso3	date	admin1	admin2	market	market_id	latitude	longitude	category	commodity	commodity_id	unit	priceflag	pricetype	currency	price
AFG	2025-01-15	Badakhshan	Argo	Badakhshan	3690	37.04	70.46	cereals and tubers	Bread	55	KG	aggregate	Retail	AFN	77
AFG	2025-01-15	Badakhshan	Argo	Badakhshan	3690	37.04	70.46	cereals and tubers	Rice (high quality)	247	KG	aggregate	Retail	AFN	88.54
AFG	2025-01-15	Badakhshan	Argo	Badakhshan	3690	37.04	70.46	cereals and tubers	Rice (low quality)	145	KG	aggregate	Retail	AFN	55
AFG	2025-01-15	Badakhshan	Argo	Badakhshan	3690	37.04	70.46	cereals and tubers	Wheat	84	KG	aggregate	Retail	AFN	23.19
AFG	2025-01-15	Badakhshan	Argo	Badakhshan	3690	37.04	70.46	cereals and tubers	Wheat flour (high quality)	178	KG	aggregate	Retail	AFN	30.38



Column	Description
countryiso3	ISO3 country code
date	Price observation date
admin1	Province administrative region
admin2	District administrative region
market	Specific market where price was recorded
market_id	Unique identifier for the market
latitude	Market's geographic latitude
longitude	Market's geographic longitude
category	Broad commodity category
commodity	Specific commodity name
commodity_id	Unique identifier for the commodity

Column	Description
unit	Unit of measurement (KG)
priceflag	Data provenance: actual/aggregate
pricetype	Retail (consumer price) or wholesale (distributor price)
currency	Local currency of the recorded price (e.g., IDR)
price	Price in local currency
usdprice	Price converted to USD (standardizedcomparison)

Dataset Overview

Data Preparation



Data Preparation

01

Import & Inspection

Load the WFP CSV with pandas; check shape, data types, missing values, and duplicates.

02

Consistency & Standardization

Standardize market and commodity name spelling/casing; confirm unit and currency consistency across records.

03

Data Type Conversion

The date column is converted to datetime to enable time-component extraction and regime segmentation.

04

Feature Engineering

New features are created to support analysis and modeling.

05

Monthly Aggregation

Raw price observations are aggregated to a consistent monthly series per market and commodity to stabilize the regime-detection and forecasting pipeline.



Data Quality Findings



Missing Values

Missing values in usdprice and various categorical columns. Handled by imputing 0 for usdprices and "Unknown" for others.



Duplicate Rows

0 duplicates found.



Consistent Format

All categorical columns verified consistent.

```
usdprice nulls : 3,460 (0.19%)
usdprice nulls by country:
countryiso3
SOM      3011
MDA       449
Name: count, dtype: int64
```

```
admin1 nulls      : 1,490 (0.08%)
admin1 nulls by country:
countryiso3
NIC       956
EGY       190
TUR       173
SWZ        81
SOM        72
MWI        18
Name: count, dtype: int64
```

Global Outlier

IQR range : \$-2.2000 – \$5.4800

Outliers : 217,719 (11.78%)

Outliers by source year:

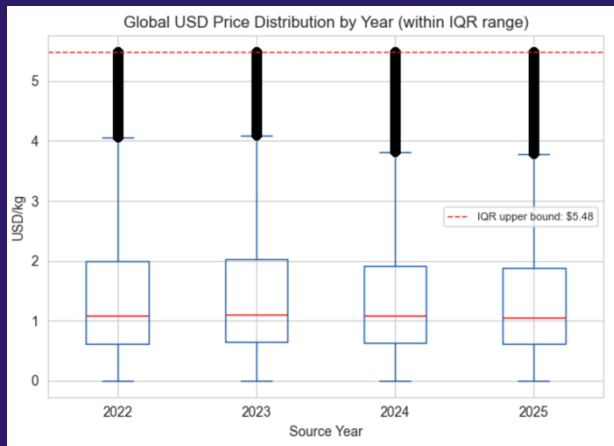
source_year

2022 59121

2023 54949

2024 51299

2025 52350



Exploratory Outlier Detection

The IQR method is applied **globally first** (across all countries and commodities) to understand overall data quality, using standard IQR bounds ($Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$).

Outlier range %:

11,78% of global rows flagged

Decision

Outliers are flagged, **not removed**.

Reason

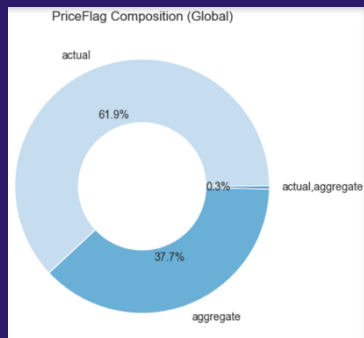
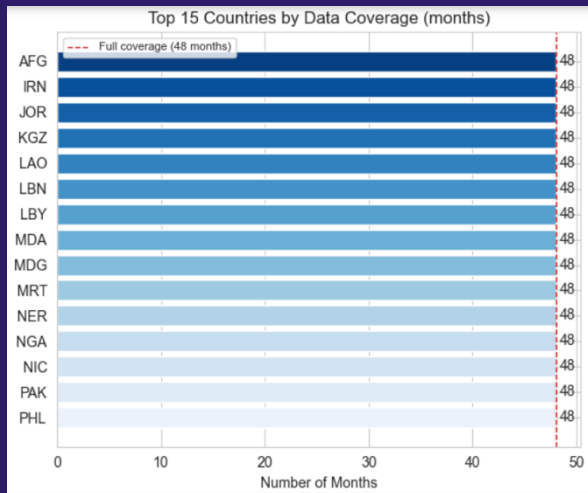
These outliers represent genuine **price extremes across different countries and commodities**, not data entry errors. Removing them would distort cross-country comparisons and hide legitimate market events.

Exploratory Data Analysis

Global

Indonesian deep-dive

Global Overview



Global Data Coverage

Data coverage and observation type were examined across all countries, since inconsistent coverage or a high share of estimated (non-actual) data would undermine reliability at the country level.

Coverage & Priceflag finding

Top 15 countries **show full 48-month coverage**. The complete training window this analysis requires. Countries below this threshold were deprioritized during country selection.

61.9% of global observations are actual (directly field-observed), 37.7% are aggregate (WFP-estimated), and 0.3% are a mix of both.

Why it matters

Full coverage and a healthy share of directly-observed data are two separate quality dimensions. A country could have complete monthly coverage but be entirely aggregate (estimated), or vice versa.

Outlier by Country

```
Total outliers flagged : 171,965
Outlier rate           : 9.31%
```

```
Top 10 countries by outlier count:
countryiso3
```

```
IDN      20835
SYR      12542
MLI       8268
GNB       6330
KHM       6273
GMB       6245
UKR       6167
BEN       6085
CMR       5838
TCD       4283
dtype: int64
```

Outlier Detection

The IQR method is **grouped by countries** (across all commodities), using standard IQR bounds ($Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$)

Results

171,965 outliers flagged (9,31% outlier rate)

By Country

Indonesia (IDN) sits at the top of the list with 20,835 flagged outliers nearly double Syria's 12,542 (2nd), suggesting an unusually wide gap in commodity pricing that makes it worth further investigation.

Decision

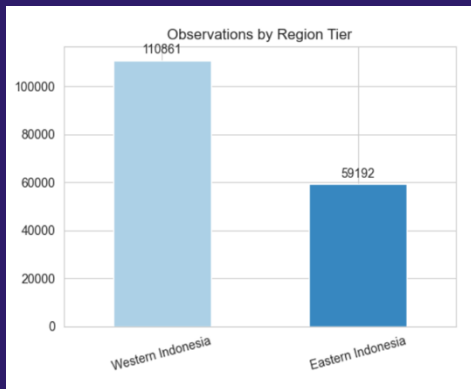
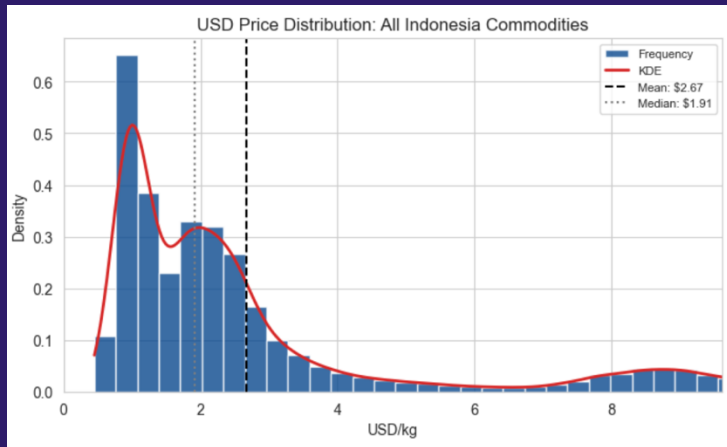
Outliers are kept, **not removed**.

Reason

Outliers represent genuine price extremes, not data entry errors. Removing them would hide the legitimate market events.

Indonesia had the highest absolute number of flagged observations. Because countries contain different numbers of records, this was treated as an exploratory signal.

Indonesia's Commodity Overview



Distribution Analysis

Price distributions were examined across Indonesia's commodities using histogram.

Finding

Distribution is **right-skewed** with a small number of high-priced commodities (primarily beef) pulling the overall average upward. It means most commodities **sit at the lower end of pricing**, with a few more expensive commodities.

Technical Consequences

Machine learning models would likely **capture the lower side while underestimating the minority**.

Regional Data Density

Observation counts are uneven across Indonesia: Western Indonesia accounts for 110,861 observations vs. Eastern Indonesia's 59,192 (roughly a 2:1 ratio). This reflects **Western Indonesia's denser market network** (more provinces, more populous regions) rather than a data quality issue.

Indonesia's Commodity Overview

34
Provinces

9
Commodity

220
Market

Regional Price Gap



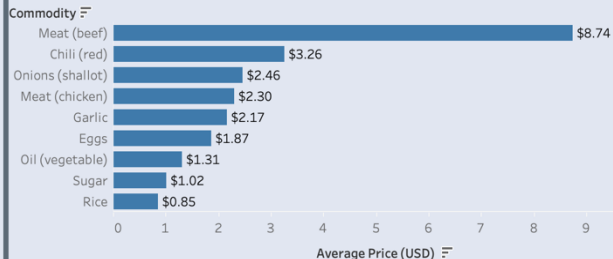
Regional Price Gap

Eastern Indonesia averages \$2.55, roughly 6% higher than Western Indonesia's \$2.40, a pattern consistent across the commodity basket, not isolated to one product. This reflects longer supply chains and higher logistics costs across Indonesia's eastern archipelago.

Commodity Price Rank

This rank reflects more of **pricing gaps across commodities** in Indonesia where Beef sits in a tier of its own, priced over 2.5x higher than the next commodity (Chili at \$3.26).

Commodity Price Rank



Indonesia's Commodity Overview

Commodity Price Volatility

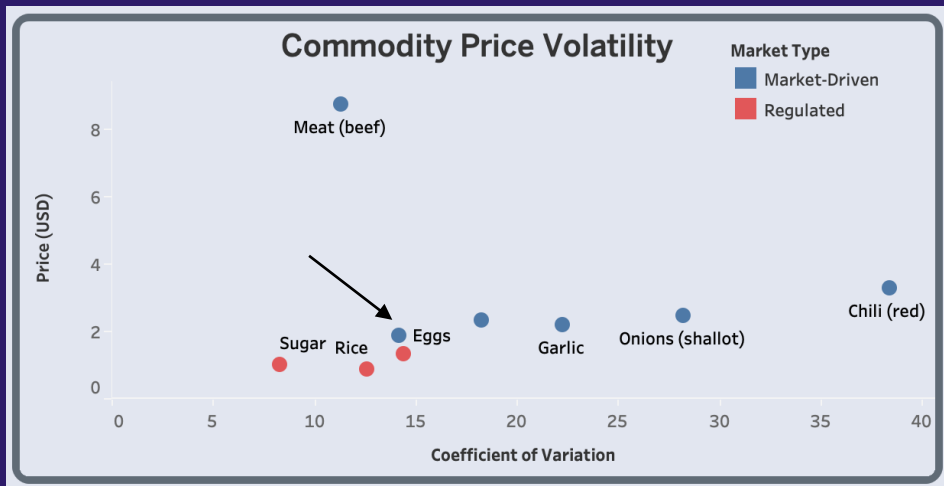
Volatility (Coefficient of Variation) **varies dramatically** across Indonesia's commodity basket. Eggs sit at a notably low CV (~14%, less than 20%) among market-driven commodities, positioned **close to the regulated cluster despite having no government price intervention**.

Why this matters

Regulated commodities (Sugar, Rice, Oil) show artificially suppressed volatility due to price controls, not genuine market stability.

Among the unregulated market-driven commodities (blue), eggs are the most stable. Meat (beef) is generally seen as a premium commodity due to its high price.

Therefore, this combination (low volatility without price regulation) makes eggs the cleanest available signal for further analysis in this dataset.



Variable Selection

Criterion	Rice	Sugar	Oil	Garlic	Onions	Chilli	Chicken	Beef	Eggs
No price regulation	✗	✗	✗	✓	✓	✓	✓	✓	✓
Low volatility	✓	✓	✓	✗	✗	✗	✓	✓	✓
Broad consumption	✓	✓	✓	✓	✓	✓	✓	✗	✓
Selected									✓

Selection Summary

Indonesia is selected as the country of focus with 220 markets across 34 provinces, and clear **food security relevance** with a notably **high global outlier** count (the highest of any country) further signaling a **market with frequent, genuine price volatility worth investigating in depth**.

Eggs are the only commodity **passing all three selection criteria simultaneously**: no government price intervention, low volatility, and broad consumption across income levels. This combination makes eggs the most methodologically sound choice for further analysis.

Chicken meets similar criteria and was a close second candidate. Eggs were ultimately selected for their marginally lower volatility and broader relevance, relatively affordable and widely consumed protein but chicken would be a reasonable extension for future analysis.

With both country and commodity formally selected, **the remainder of this analysis focuses exclusively on Indonesia's egg market**.

Analysis Phase

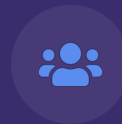
Indonesia's Egg Commodity

Objectives



Understand Price Behavior

Identify any price patterns in Indonesia's egg market from the WFP dataset.



Forecasting

Build and compare forecasting models to find the most reliable one for egg prices.



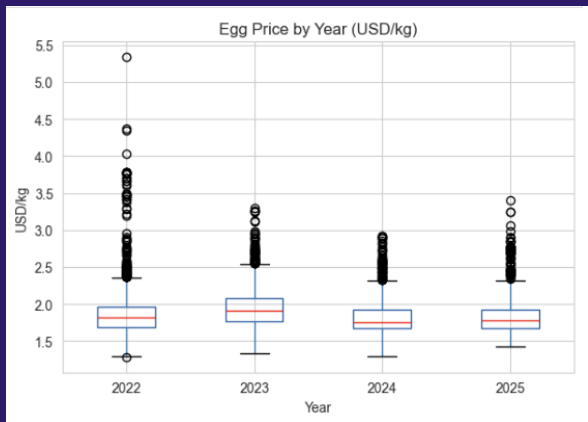
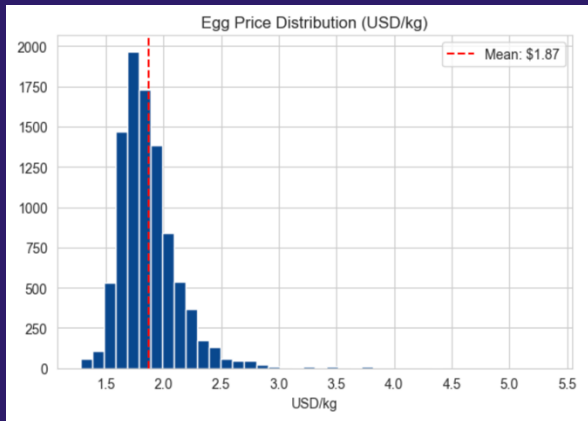
Deliver Data-Driven Recommendations

Translate price patterns and regime shifts into recommendations for commercial and policy use.

Key Feature Engineering

Feature Name	Formula	Notes
year, month	<code>df['date'].dt.year / .month</code>	Time components for trend and seasonal analysis
is_outlier	Per-country IQR method on usdprice	Flags statistical outliers
region_tier	Province mapped to Eastern/Western Indonesia	Enables regional price comparison

Indonesia's Egg Price Distribution



Price Distribution

Egg prices are **right-skewed**, signals **occasional sharp price spikes** rather than a normal spread. A first hint that the **market has periodic shocks layered on a stable baseline**.

Outliers

The typical price stayed steady from 2022 to 2025, but every single year had a few of unusually high spikes. 2022 saw the most extreme jumps, with prices reaching as high as \$5.5/kg in some markets.

By 2024–2025, those spikes were still happening, but they weren't nearly as dramatic. It's a sign that the market's price shocks have been calming down.

Why it matters

This uneven pattern matters for how the forecast is built. Treating egg prices as if they follow a simple, even curve would underestimate how sharp a price spike can get. And a plain trend line wouldn't pick up on something important: each year's shocks have been getting a little less extreme than the last. It **reinforces the case for a regime-aware model that separates the stable baseline from transient shocks**.

Indonesia's Egg Market Overview



Regional distribution

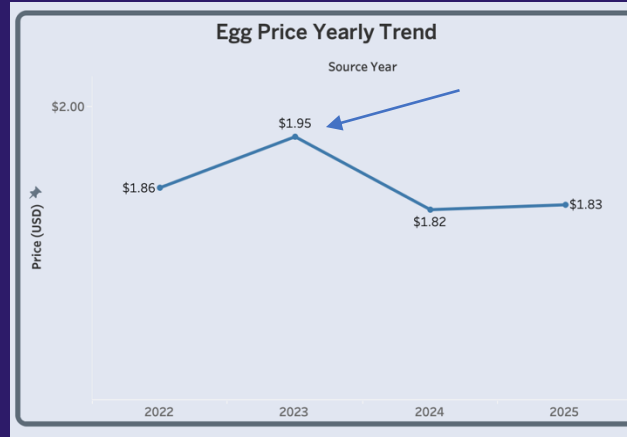
Average prices vary meaningfully across Indonesia's markets. Coverage spans Sumatra, Java, Kalimantan, Sulawesi, and Papua, with **denser clusters in western Indonesia** and a sparser, comparatively **higher-priced set of markets further east**. This may be reflected by different local production levels, market concentration and infrastructure in this eastern region.

Why it matters

A market in Java and a market in Papua might be **paying very different prices for eggs on the exact same day**. For a business, that means relying on the national average alone could mean paying too much or too little, depending on where the eggs come from.

For food security, the places paying the most for eggs (mostly in eastern Indonesia) are also the farthest from the supply routes that could bring prices down.

Egg's Price Trend



Overview

Egg prices in Indonesia followed phases between 2022 and 2025:

- **Gradual rise** (2022 - 2023): prices climbed steadily to a peak of \$1.95/kg
- **Sharp correction** (2023 - 2024): prices fell back down as the market absorbed the shock
- **Stabilization phase** (2024 – 2025): prices stabilized post crisis

The **2023 peak coincides with a corn feed cost crisis**. Corn is the primary input for poultry feed in Indonesia, and elevated global corn prices **combined with El Niño-driven supply pressure** pushed egg production costs sharply upward.

This isn't random noise. It's a **genuine structural break**. A model that doesn't account for this shift risks either overreacting to the spike (assuming prices will keep rising) or underreacting to the new stable baseline. **This motivates the regime-aware approach** covered later.

Source

All About Feed (2024).
Indonesia: Poultry industry crisis triggers feed market turbulence. [LINK](#)

Split Ratio & Stationarity Test

Split	Train Months	Test Months	Train End	Test Start	Seasonal	MAPE
70/30	33	15	2024-09	2024-10	(0, 0, 0, 12)	2,21%
75/25	36	12	2024-12	2025-01	(0, 0, 0, 12)	5,53%
80/20	38	10	2025-02	2025-03	(0, 0, 0, 12)	2,64%
85/15	40	8	2025-04	2025-05	(0, 0, 1, 12)	4,22%
90/10	43	5	2025-07	2025-08	(0, 0, 1, 12)	2,40%

Split Ratio Selection Summary

70/30, 75/25, and 80/20 **all settled into a non-seasonal (0,0,0,12) fit**. Their MAPE looks competitive, but a model that dropped seasonality means it stopped trying to capture the yearly egg price cycle at all. Only two splits (85/15 and 90/10) kept the seasonal pattern in the model. 90/10 shows the lowest error (2.40%), but **its test window is only 5 months long**. 6 months is the minimum needed for a fair test, so that low number is more likely luck than a true reflection of how well the model performs.

Selected: 85/15

40 training months and an 8-month test window is long enough to hold up and still confirms a genuine seasonal SARIMA fit. Its 4.22% MAPE is higher than 90/10's, but it's the honest number, earned by proving seasonality rather than by a lucky window.

Stationarity Test

ADF test on the training data: $p = 0.0436$ which shows borderline **stationary**.

2023 shock motivated regime-aware modeling over a single fixed-parameter approach.

Model Comparison

Model	MAPE	Notes
Baseline (MA-3)	1,63%	Simple moving-average benchmark
SES	2,24%	Best simple statistical model. Tracks recent price level closely
SARIMA (standard)	4,22%	Fits one pattern to the whole series, missing the 2023 regime shift
Holt-Winters	4,41%	Seasonality zeroed out
Prophet	4,96%	Overfitted seasonality
Holt-Winters Trend	8,95%	Trend overshoot, no seasonal component
Regime-aware Adaptive SARIMA	1,65%	Adapts to abnormality/regime shifts

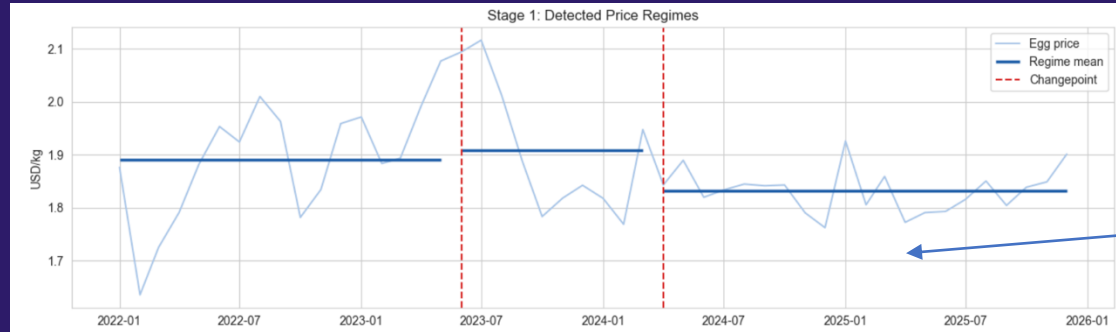
Model Selection Summary

The baseline achieved the lowest MAPE at 1.63%. Regime-aware adaptive SARIMA performed nearly as well at 1.65% and was **selected for its structural awareness** rather than superior point accuracy.

Every standard statistical model (SARIMA, Holt-Winters, Prophet, Holt's Linear) performed worse than Baseline because none of them account for the regime shift caused by the 2023 feed cost crisis.

Regime-Aware Adaptive SARIMA closes that gap by detecting and adapting to the structural breaks in the data rather than assuming one fixed pattern across the entire series. It is selected not because it beats Baseline outright, but because it's the only model that combines Baseline-level accuracy with genuine seasonal and structural awareness, making it the more reliable choice for forecasting beyond the training window.

Advanced Modeling : Regime Aware Forecasting



Model	MAPE
Static	4,22%
Adaptive	1,65%
Error reduction	~61%

Regime Detection

The price series was segmented into three regimes based on two structural breaks corresponding to the corn feed crisis and post-crisis normalization.

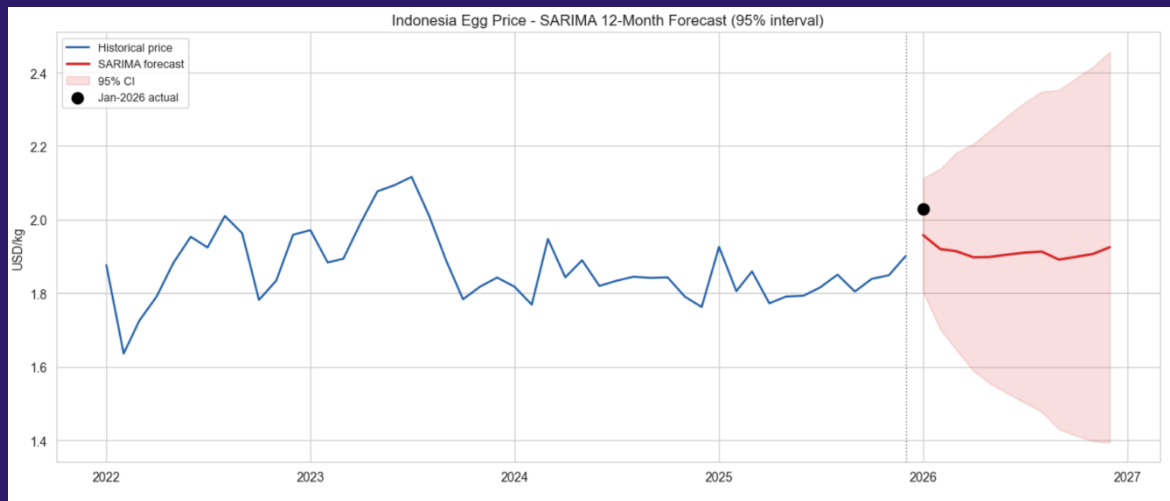
Each regime shows a distinct mean price level, confirming the market didn't move on one continuous trend, but **shifted between discrete phases**.

Changepoints were detected using training data only, while the test period remained unseen until evaluation.

Adaptive Model Limitation

The **adaptive model doesn't predict a break before it happens**, no model can. Its advantage comes from correcting faster once new data confirms a shift has occurred, rather than staying anchored to an outdated regime. This is the operational case for **walk-forward forecasting**: refit monthly as new prices are published, rather than committing to a single static model.

Final Forecast (Using Regime Aware Approach)



12-Month Forecast

The final regime-aware SARIMA model projects egg prices through **January to December 2026**.

Uncertainty Widens the Horizon

The 95% confidence interval starts narrow in January and widens substantially by December. A mathematically expected property of forecasting further from the last known data point, and a genuine signal: **confidence in month 1 is much higher than confidence in month 12.**

January 2026: Forecast vs Actual

The gap is explained by **Ramadan falling in February 2026**, the earliest date in the training dataset's history, pulling pre-Ramadan demand into January for the first time.

No model trained on 2022–2025 history had precedent for this specific lunar calendar shift, but the model's uncertainty band was wide enough to still capture the actual outcome.

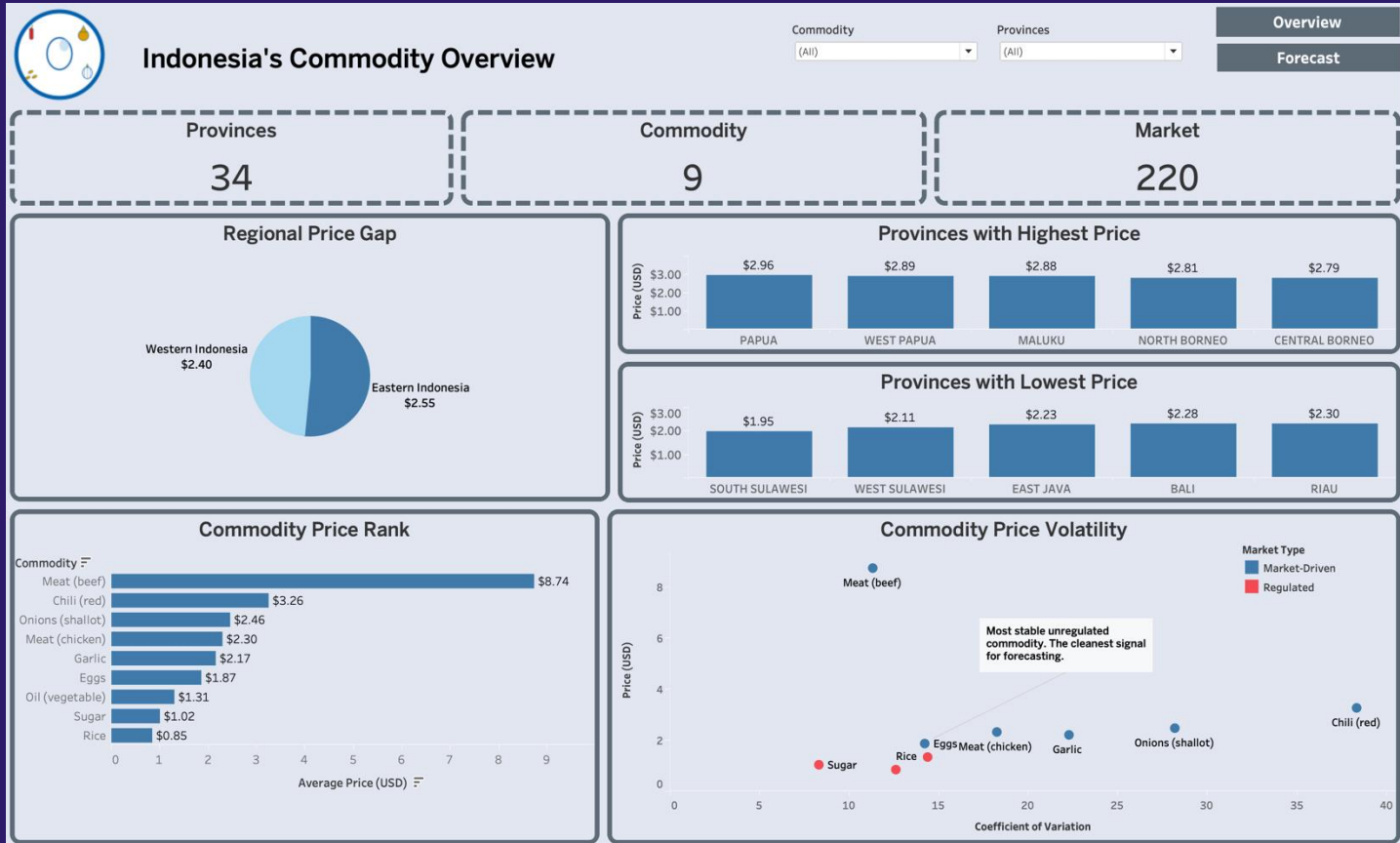
Dashboard Outputs

2 Interactive Dashboards · Tableau Public

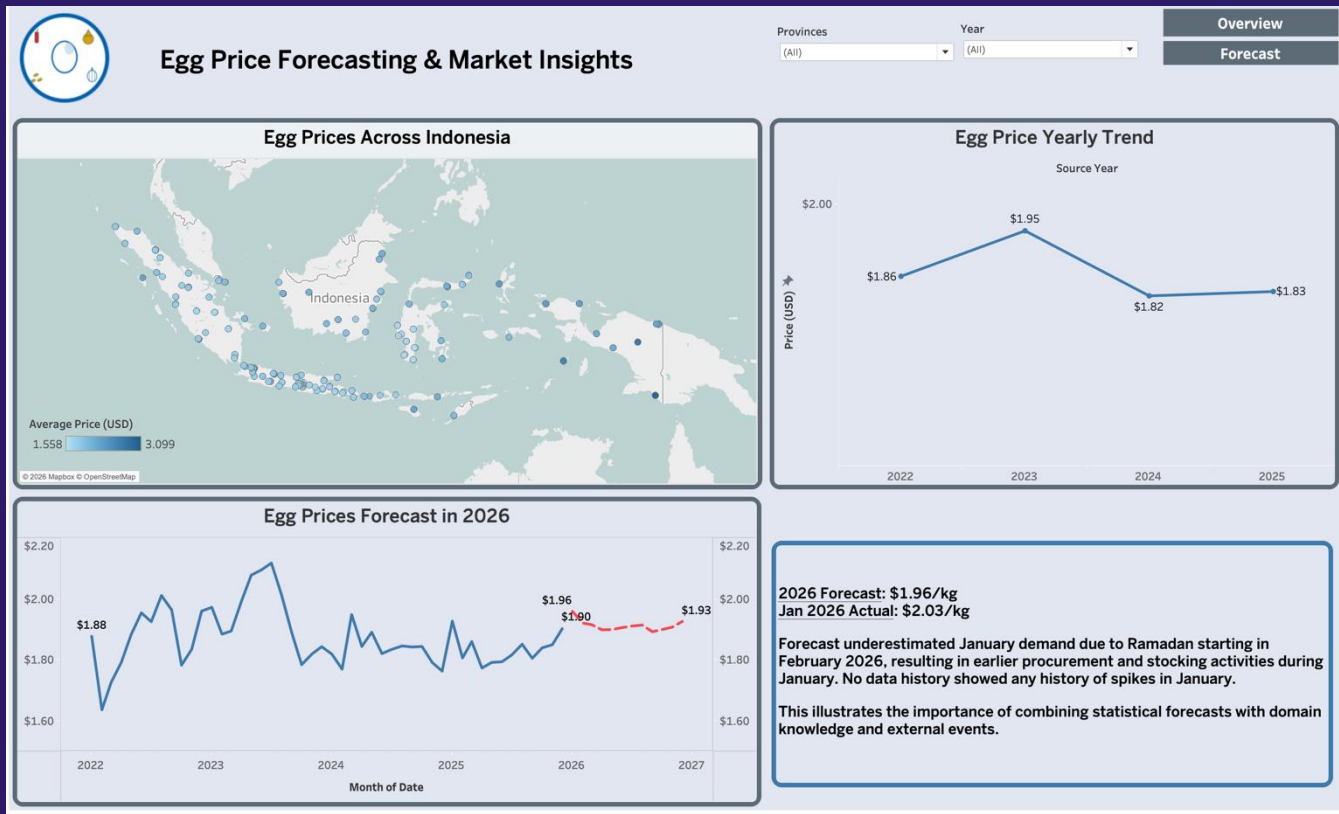
**Indonesia's
Commodity Overview**

Forecast

Dashboard 1: Indonesia's Commodity Overview



Dashboard 2: Egg Forecast



Note: the dashboard version displays the point forecast only for readability. The full model output, including the 95% confidence interval already shown earlier.

Conclusions & Recommendations

Egg prices in Indonesia **moved through two very different chapters over the past four years**: a real market shock in 2023 that no model saw coming, followed by a calmer stretch where fancy forecasting tools didn't add much value over simply looking at recent prices.

After testing different approaches, the clearest lesson wasn't which one "won". It's that the right tool depends on the moment. **Simple methods work fine when the market is calm**, but a smarter, adaptive approach pays off specifically when things are shifting, reducing error by approximately 61% compared with static SARIMA during the 2023–2024 transition.

The January 2026 check-in taught a similar lesson. The range of possible outcomes the model predicted still covered what happened. **The gap came down to Ramadan falling earlier than it ever had in the four years of data** used to build the model, pulling holiday demand into a month that had never seen a spike before. The point forecast missed the January increase, although the actual price remained within the uncertainty level.

That's exactly why **statistics alone isn't enough here**. A model can learn from the past, but it can't know what it's never been shown. Recognizing that gap, and **stepping in with real-world judgment when it appears, is still a human job**.

Commercial Recommendation

01

Use the forecast as an early-warning tool for inventory planning, particularly before predictable high-demand periods. Build a buffer stock around those periods each year. (like Ramadan, Christmas, etc.)

Regional retailers across the Gulf already apply this principle, pre-positioning inventory weeks ahead of Ramadan rather than reacting to shortages after they occur (The Media Line, 2026).

Policy Recommendation

02

Extend this regime-detection approach to other unregulated staples. Government agencies can combine the forecast, uncertainty range and regional price differences with current supply information to identify where intervention may be needed, such as coordinating distribution, monitoring potential shortages or preparing market-stabilization measures.

This direction is backed by emerging research: a 2026 Indonesian study on multi-commodity food price forecasting ahead of Eid Al-Fitr independently confirms that Indonesian food prices require dedicated, calendar-aware forecasting around Ramadan (Murtani, Kobi, & Imelda, 2026).

Reflection

01 What this dataset cannot answer

With only 4 years of data and just one major price shock, we can't cleanly tell apart "this happened because of Ramadan" from "this happened because of the 2023 feed crisis" as they overlapped in the only window we had. We'd need several more years of data to separate the two properly.

02 What changed during the analysis process

We started out trying to build the most advanced model possible. A failed attempt to bake Ramadan directly into the model (which made things much worse) also pushed the final answer toward a human-in-the-loop approach instead of a purely statistical one. The initial pipeline contained data leakage in changepoint detection, so the process was corrected to use training data only and all test results were recalculated.

03 What still needs to be validated?

We've only checked the forecast against one real month so far (January 2026). More months are needed to know if this was a one-off or a pattern. It's also worth checking whether other everyday foods (like chicken), not just eggs, show the same Ramadan-driven blind spot.

Q&A